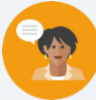


1.8 Line Segments

Lesson Introduction

	Benchmark	MA.912.GR.5.1 Construct a copy of a segment or an angle.			Learning Targets	- I can construct a copy of a line segment. - I can use the segment addition postulate.
	Benchmark Coherence	Building On			Working Toward	
		N/A			N/A	
	Essential Questions	- How can I make a copy of a line segment? - How are the lengths of pieces of a line segment related to the entire line segment? - How can I use the segment addition postulate to find the length of a line segment?				
	Lesson Narrative	Students are guided on how to make a copy of a line segment using a ruler, patty paper, compass, and toothpick. Using the ruler postulate, definition of collinear points, and the segment addition postulate students solve problems to determine the length of a line segment. Students work through the remainder of the lesson applying the segment addition postulate in mathematical and real-world applications.				
	Component Summary	1.8.1 Warm-Up (~5 min.)	Using critical thinking to find area (<i>SE p. 53</i>)	1.8.3 Guided Practice (10 min.)	Apply the segment addition postulate to solve mathematical problems (<i>SE p. 56</i>)	
		1.8.2 Guided Instruction (25 min.)	Copying line segments and developing the segment addition postulate (<i>SE p. 54</i>)	1.8.4 Try It! (5 min.)	Apply the segment addition postulate to solve a real-world problem (<i>SE p. 57</i>)	
		1.8.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 58</i>)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Ruler Postulate - Collinear Points - Segment Addition Postulate <u>Additional Terms:</u> N/A - Line - Line Segment - Ray		- Ruler - Compass - Patty Paper - Toothpicks (Optional) Graph Paper		N/A

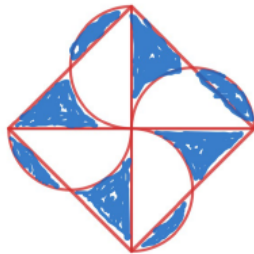
 Teacher Tips	Common Misconceptions	Things to Consider
	Students may swap the notation for a line segment and the length of the line segment by leaving off or including the segment symbol inappropriately.	- In 1.8.2 Guided Instruction, before introducing the Ruler Postulate, consider engaging in a quick number talk to help students recall how to find the distance between two points on a number line. Taking a moment here will help students to better understand the formal definition of the Ruler Postulate and answer question 5c in the same component. - If time is an issue, then consider assigning 1.8.4 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect In 1.8.2 Guided Instruction question 3, students are using a compass to copy a line segment. Students will choose where to place point M on their arc. Engage students in conversation of their placement of M , comparing their placements and connecting why all of their placements, as long as they are on the arc, are correct. Encourage the use of the appropriate vocabulary: circle, radius, center, and distance.	MA.K12.MTR.2.1: Multiple Representations Students build an understanding using constructions and express connections between the concepts and representations. In 1.8.2 Guided Instruction students copy line segments using a ruler, patty paper, compass, and toothpick. Students relate the measurement of a line segment using the Ruler Postulate and then apply the measurement using the Segment Addition Postulate.
	1.8.2 Guided Instruction question 4 provides students with a kinesthetic entry-point with the material. Using toothpicks, students can demonstrate a specific transformation. This activity will help students who need tactile stimulation digest the concepts.	
Lesson Notes:		

1.8.1 Warm-Up: Semicircles and Triangles

Component Overview: Students apply critical thinking to solve a question about the area of an image.



1.8.1 Warm-Up: Semicircles and Triangles



- What do you notice about the picture?
Answers will vary. Some is shaded, some is not. It has half circles.
- If the radius of each semicircle is 5, then what is the length of each of the square's diagonals?
20
- Dina determined the area of the shaded region using the drawing shown. Find the total shaded area.
 $A = 100 \text{ units}^2$



For students who are stuck remind them that the radius is the distance from the middle of the circle to the edge. This diagram tends to “trick” the eye by making it appear that the distance from the center of the square to the corner is the radius; however, this is the diameter. There are 4 iterations of the radius across the long diagonal.

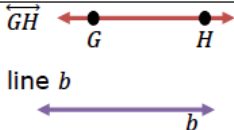
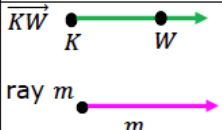
Allow students to struggle or to make mistakes. Depending on your class routine this may be a good problem for students to work on together. Limit the time students spend on the problem to about 5 minutes. Let students know that the expectation is to try and to think through the question and that it is okay to get it wrong.

1.8.2 Guided Instruction: Line Segments

Component Overview: Students review the terms line, line segment, and ray as well as how to name them. Next, students make a copy of a line segment using a ruler, patty paper, compass, and toothpick. The ruler postulate and collinear points are presented to develop the segment addition postulate. Students use the segment addition postulate to solve for a variable and for different segment lengths.



1.8.2 Guided Instruction: Line Segments

Line	Line Segment	Ray
A line is an undefined term.	A part of a line that is bounded by two distinct end points.	A part of a line that has a fixed starting point and extends infinitely in one direction.
		

When naming a line segment, the line segment symbol should be used; however, when giving the length of a line segment, the symbol should not be used.

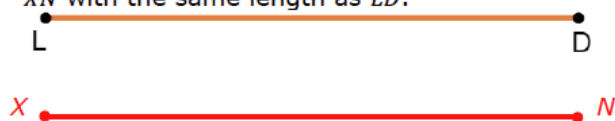
$$\overline{TM} \quad TM = 12 \text{ millimeters}$$

1. $\overline{RK}$ is shown. Use a ruler to measure the line segment in centimeters.



Note: Actual length will vary depending on student lesson format.

2. Use patty paper to copy $\overline{LD}$. Then, draw line segment $\overline{XN}$ with the same length as $\overline{LD}$.



Students may be familiar with these terms and representations. The formal introduction and correct naming of line, line segment, and ray is a fundamental part of geometry.

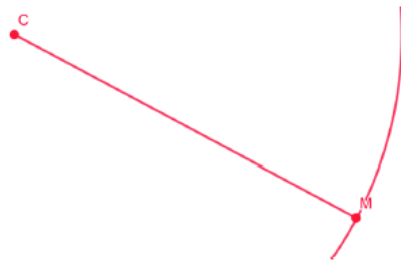


Before moving to question 1, consider asking the following:

- What are some ways that you can copy a line segment?
- How accurate are those ways?
- What tools can help copy a line segment?

1.8.2 Guided Instruction: Line Segments (continued)

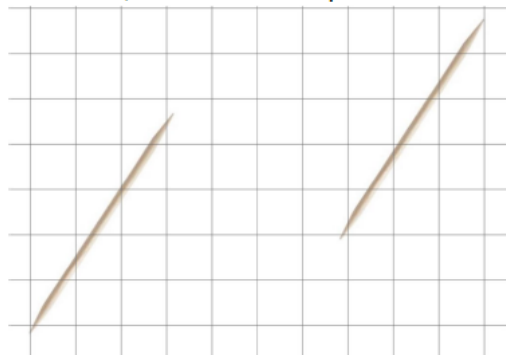
3. Follow these steps to draw a copy of $\overline{LD}$ using a compass.
 - Draw point C . This will be one endpoint of the new line segment.
 - Set the compass' point on point L .
 - Adjust the compass' width to point D so the compass' width is equal to the length of $\overline{LD}$.
 - Without changing the compass' width, place the compass' point on C and draw an arc roughly where the other endpoint will be.
 - Pick a point M on the arc that will be the other endpoint of the new line segment.
 - Draw line segment $\overline{CM}$ using a straightedge.



Engage students in the Literacy and Language Routine Compare and Connect to discuss the placement of point M . Encourage the correct use of vocabulary, including circle, radius, center, and distance.

1.8.2 Guided Instruction: Line Segments (continued)

4. Discuss with your partner how to show that the two toothpicks are congruent using each method in the table. Then, write a brief explanation for each method.



Transformation	<i>Translate each point of the left toothpick to the right seven units and up two units.</i>
Patty Paper	<i>Copy either toothpick onto the Patty Paper. Move the Patty Paper to cover the other toothpick.</i>
Compass	<i>Set the point of the compass on one end of a toothpick and open the compass to reach the other end of the same toothpick. Move the point on the compass to the end of the second toothpick and draw an arc. The toothpicks are congruent if the arc passes through the second endpoint of the second toothpick.</i>



Ruler Postulate

The points of a line can be placed in a one-to-one correspondence with the real numbers such that the distance between two distinct points is the absolute value of the difference of the corresponding real numbers.



Students may benefit from using toothpicks to explore transformations. For a kinesthetic approach, we are connecting using the compass and using a toothpick.



Students have been working with number lines since elementary school. Discuss that the concept of the ruler postulate is nothing new for them, just a formalized explanation.

1.8.2 Guided Instruction: Line Segments (continued)

5. $\overline{KC}$ and $\overline{YC}$ are shown.



- a. Discuss with your partner how aligning a ruler with $\overline{KC}$ illustrates the ruler postulate. Summarize your discussion.

Answers will vary. Place a ruler beside the line so that the points line up with positions on the ruler. Subtract the corresponding number on the ruler for one point from the corresponding number on the ruler for the other point. The absolute value of this difference is the length of the segment between the two points.

- b. Use a ruler to measure each line segment. Include units.

$KY = 4.4 \text{ cm}$ $YC = 6.5 \text{ cm}$ $KC = 10.9 \text{ cm}$
Actual length may vary based on units of measure.

- c. Discuss with your partner why $KY + YC = KC$ should be true.

Answers will vary. The line is equal to the sum of its parts.

Points on the same line are called **collinear points**.



Segment Addition Postulate

If points A , X , and B are collinear points and point X is between points A and B , then $AX + XB = AB$.

6. $\overline{KW}$ and $\overline{WT}$ are shown, where $KW = 4x - 7$, $WT = 23.5$, and $KT = 7x - 3$.



- a. Determine the value of x .

$x = 6.5$

- b. What is the length of $\overline{KW}$?

$KW = 19$



To provide relevance to the students, lead a discussion of where the segment addition postulate may be useful in the real world. Consider a variety of professions that may use line segments, like architecture or graphic design. Consider manipulatives, like Lego and K'nex as places they may have used line segment addition in the past.



Monitor students as they solve the equation in question 6. Note students who appear struggling. Consider utilizing a small group during 1.8.3 Guided Practice.

1.8.3 Guided Practice: Segment Addition

Component Overview: Students will apply the segment addition postulate and solve for variables and segment lengths.



1.8.3 Guided Practice: Segment Addition

1. $\overline{YB}$, $\overline{BW}$, and $\overline{WH}$ are shown, where $YB = 2x$, $BW = x + 4$, $WH = 3x - 1$, and $BH = 35$.



- Determine the value of x .
 $x = 8$
 - What is the length of $\overline{YW}$?
 $YW = 28$
2. Points K , L , M , and N are collinear such that $KM = LN$.

- Draw a diagram that represents points K , L , M , and N .

Answers will vary.



- Compare your diagram with your partner. Discuss whether or not the diagrams drawn must all be the same. Summarize your discussion.

Answers will vary. No, they do not have to be the same. As long as all four points are collinear and $KM = LN$, they could be placed anywhere along the line.

- Determine the value of x if $KM = 6x - 6.2$ and $LN = 5x - 2$.
 $x = 4.2$
- Using the information from c, determine the length of $\overline{KM}$.
 $KM = 19$



Continue to monitor students as they solve equations. Consider directing students to the Student Area online for a refresher on solving equations.



Question 2 leaves students to test out the placement of points when not given an image and may try labeling the points in a variety of ways. Because of the declaration that $\overline{KM} = \overline{LN}$, then the equations are to be set equal, and the answer for the variable will not change.

1.8.4 Try It: Ocean to Lake Florida Trail

Component Overview: Students will use the segment addition postulate to solve a real-world problem.



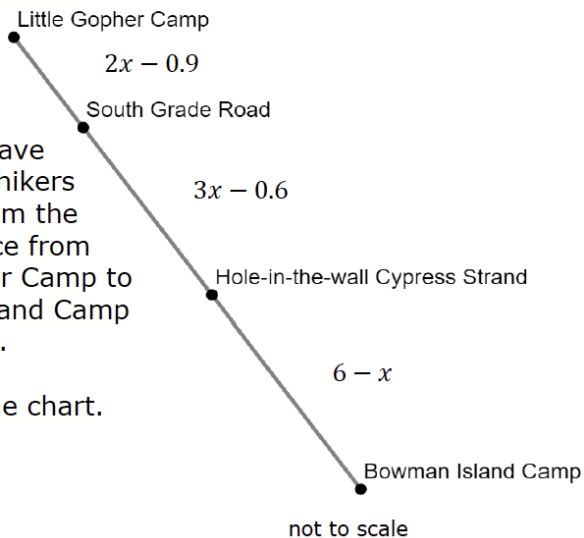
1.8.4 Try It: Ocean to Lake Florida Trail

- The Ocean to Lake Hiking Trail runs for approximately 63 miles, west from Hobe Sound Beach to Lake Okeechobee. Leigh Ann created a simple map of part of the trail that goes through J.W. Corbett Wildlife Management Area.

Her map is shown.

Leigh Ann gave the map to hikers and told them the total distance from Little Gopher Camp to Bowman Island Camp is 10.5 miles.

Complete the chart.



Consider showing students a map of the Ocean to Lake hiking trail and discuss why a simple map may be useful.



Students can assess and evaluate the reasonableness of their solutions, MA.K12.MTR.6.1, by substituting their answers in for all lengths in the problem. If the smaller line segment lengths sum to equal the entire length, then students can feel confident that their answer is correct.

Hike	Distance (miles)
Little Gopher Camp to South Grade Road	2.1 miles
South Grade Road to Hole-in-the-wall Cypress Strand	3.9 miles
Hole-in-the-wall Cypress Strand to Bowman Island Camp	4.5 miles

1.8.5 Wrap Up: Cool Down

Component Overview: *Students create an example demonstrating the Segment Addition Postulate.*



1.8.5 Wrap-Up: Cool Down

1. Create an example of the Segment Addition Postulate and show the steps to finding the total length.

Answers will vary.



Student responses can be used to identify misconceptions and plan for upcoming instruction.